ITA03-MOBILE COMPUTING

ASSIGNMENT 3

ASSIGNMENT 3 – CO3

CO3 Coverage: Introduction to wireless networks – mobile IP - issues in mobile IP differences between cellular and ad hoc wireless networks- applications - routing protocols – Proactive & reactive Routing Protocols.

Mode: Pair Work – 2 Students per Group

Duration: 1 Week

Total Marks: 50

NAME:K.BALAJI

REG.NO:192324180

Pair-Work Principle

Both students are responsible for understanding the complete assignment.

However, each question contains clearly identified individual responsibilities so that the contribution and understanding of each student can be independently verified.

Student A and Student B must not divide the assignment question-wise. Both students must participate in every question.

The individual responsibilities intentionally alternate across questions so that both students demonstrate evidence of:

Concept Understanding → Network Analysis → Protocol Desigs→ Implementation/Simulation → Evaluation

Q1 – Wireless Network Architecture and Mobility Analysis (6 Marks)

Part A – Introduction to Wireless Networks

1. Introduction

A wireless network is a communication system that allows devices to exchange data without using physical cables. It uses radio waves, infrared or other wireless signals to transmit information.

Wireless networks allow users to communicate while moving from one location to another.

Examples:

- Wi-Fi networks.
- Mobile communication networks (4G/5G).
- Wireless sensor networks.

- Bluetooth networks.

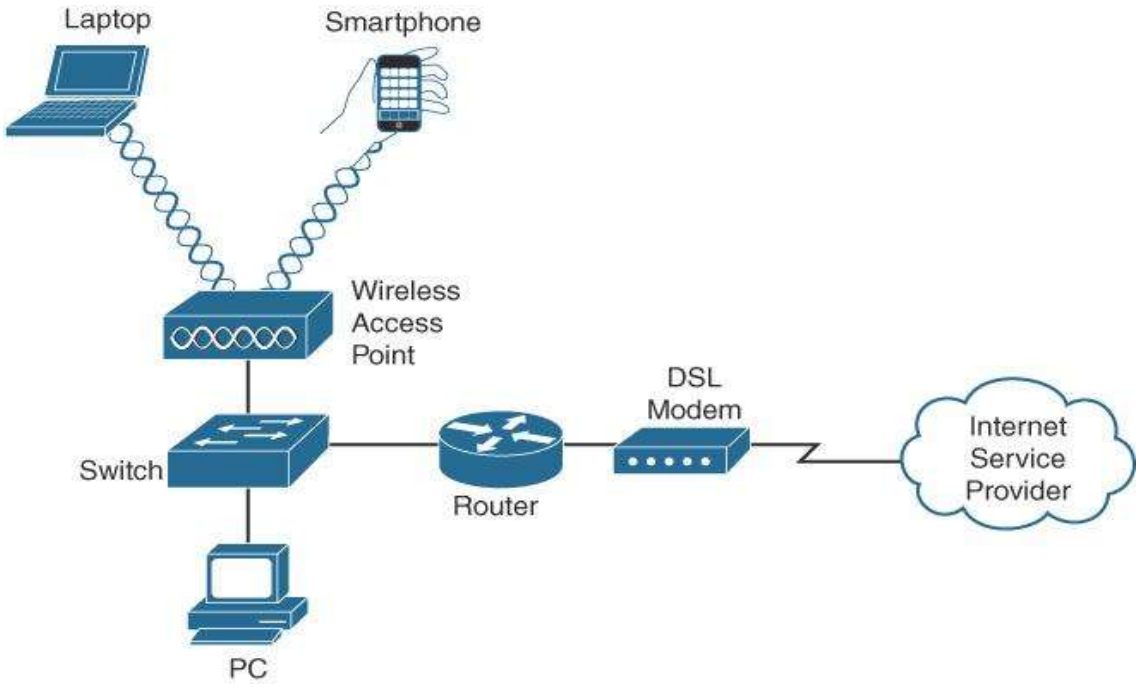
2. Characteristics of Wireless Networks

1. Mobility: Users can communicate while moving.
2. Limited Bandwidth: Wireless networks generally have limited available bandwidth.
3. Signal Interference: Radio signals may be affected by obstacles and other devices.
4. Variable Connectivity: Connections may become weak or disconnected.
5. Handoff: A mobile device can switch between different access points or base stations.
6. Security: Wireless communication requires authentication and encryption.

3. Major Components of Wireless Networks

Component	**Function**
Mobile Node	Device that sends and receives data.
Base Station	Connects mobile devices to the network.
Wireless Access Point	Provides wireless connectivity.
Router	Forwards packets between networks.
Gateway	Connects different networks.
Communication Link	Wireless connection between devices.

4. Wireless Network Architecture



Explanation of architecture:

1. Mobile devices connect to a nearby base station or access point.
2. The base station receives data from mobile devices.
3. The router forwards the packets to the destination network.
4. The gateway connects the wireless network to external networks.
5. The destination device receives the transmitted data.

5. Wireless Communication Characteristics

- Wireless communication uses radio frequency signals.
- Signal strength decreases with distance.
- Interference can reduce communication quality.
- Bandwidth is shared among connected devices.
- Mobility may cause packet loss and transmission delay.

6. Effect of Limited Bandwidth and Connectivity

Limited bandwidth reduces the amount of data that can be transmitted per second.

When many users connect to the same access point:

- Network congestion increases.
- Data transmission becomes slower.

- Packet loss may increase.
- Communication delay increases.

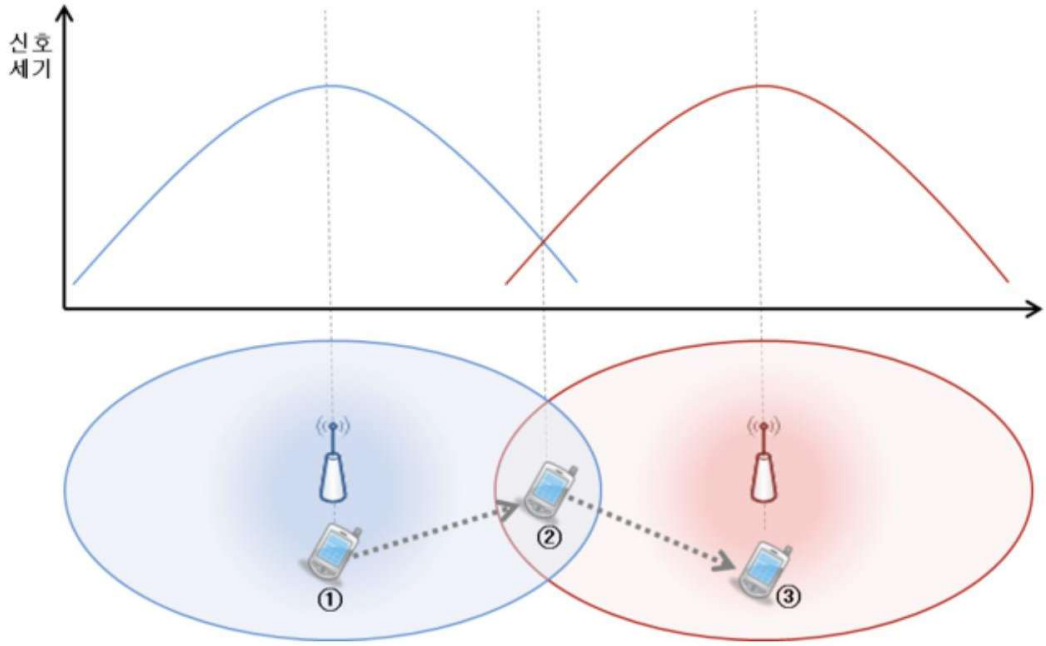
When wireless connectivity becomes weak, packets may be lost or retransmitted.

Part B – Wireless Mobility Scenario

7. Mobility Scenario

Consider a mobile user travelling from Area A to Area B.

- Area A is covered by Base Station BS1.
- Area B is covered by Base Station BS2.
- The mobile user initially connects to BS1.
- When the user moves towards Area B, the signal from BS1 becomes weak.
- The device connects to BS2 through a handoff process.



8. Handoff Process

Handoff is the process of transferring an active communication connection from one base station or access point to another without unnecessarily interrupting communication.

Steps:

1. Mobile node connects to BS1.
2. The device continuously monitors signal strength.

3. BS1 signal becomes weak.
4. The device detects BS2.
5. The network selects BS2.
6. Connection is transferred to BS2.
7. Data transmission continues through BS2.

9. Packet Loss and Transmission Delay

Packet Loss: Occurs when transmitted packets fail to reach the destination.

Transmission Delay: The time taken for a packet to travel from sender to receiver.

During handoff:

- Some packets may be lost.
- Data transmission may be temporarily interrupted.
- Delay may increase due to connection switching.
- Repeated movement may reduce communication efficiency.

10. Different Mobility Conditions

Mobility Condition	**Network Behaviour**
Low Mobility	Stable connection and low handoff frequency.
Medium Mobility	Occasional handoff and moderate delay.
High Mobility	Frequent handoff and possible packet loss.

11. Analytical Investigation – Student A

Successful Communication:

When the mobile node remains within the coverage area of BS1, data packets are transmitted successfully.

Interrupted Communication:

When the mobile node moves outside the coverage area of BS1 before connecting to BS2, communication may be interrupted.

Example observation (illustrative):

Condition	**Packet Sent**	**Packet Received**	**Packet Loss**
Stable connection	100	100	0

During handoff	100	94	6
Weak connectivity	100	85	15

Packet loss percentage:

$$Packet\ Loss = \frac{Packets\ Lost}{Packets\ Sent} \times 100$$

For handoff:

$$= \frac{6}{100} \times 100 = 6\%$$

These are sample analytical values, not measured simulation results.

12. Mobility Logic – Student B

The mobile node changes its point of attachment when the current connection becomes weak and a suitable alternative network is available.

Mobility decision sequence:

1. Monitor current signal strength.
2. Detect neighbouring networks.
3. Measure alternative signal quality.
4. Select a suitable access point.
5. Perform handoff.
6. Update network connection.
7. Resume communication.

13. Pseudocode – Wireless Mobility Management

ALGORITHM ManageWirelessMobility(MN, BS1, BS2)

INPUT:

MN – Mobile Node

BS1 – Current Base Station

BS2 – Target Base Station

OUTPUT:

Handoff status and communication status

BEGIN

```
INITIALISE CurrentBS ← BS1
INITIALISE ConnectionStatus ← CONNECTED

WHILE CommunicationRequired = TRUE DO

    Measure SignalStrength(CurrentBS)

    IF SignalStrength(CurrentBS) < MIN_SIGNAL THEN

        Detect BS2
        Measure SignalStrength(BS2)

        IF SignalStrength(BS2) >= MIN_SIGNAL THEN

            SWITCH connection TO BS2
            CurrentBS ← BS2
            ConnectionStatus ← CONNECTED

        ELSE

            ConnectionStatus ← DISCONNECTED
            DROP packets if transmission is unavailable

        END IF

    ELSE

        ConnectionStatus ← CONNECTED
```

END IF

END WHILE

RETURN ConnectionStatus

END

14. Result and Conclusion

Wireless networks provide communication without physical cables and support mobile users. However, limited bandwidth, signal interference, packet loss and connectivity interruptions affect performance.

Handoff helps maintain communication when a mobile user moves between coverage areas. Efficient mobility management reduces delay and packet loss and improves communication reliability.

Q2 – Mobile IP and Mobility Management (7 Marks)

1. Introduction to Mobile IP

Mobile IP is a network protocol that allows a mobile device to move from one network to another while maintaining its permanent IP address.

It supports continuous communication even when the mobile node changes its network location.

Example: A user travelling from Chennai to Bangalore can continue accessing network services without changing the permanent home IP address.

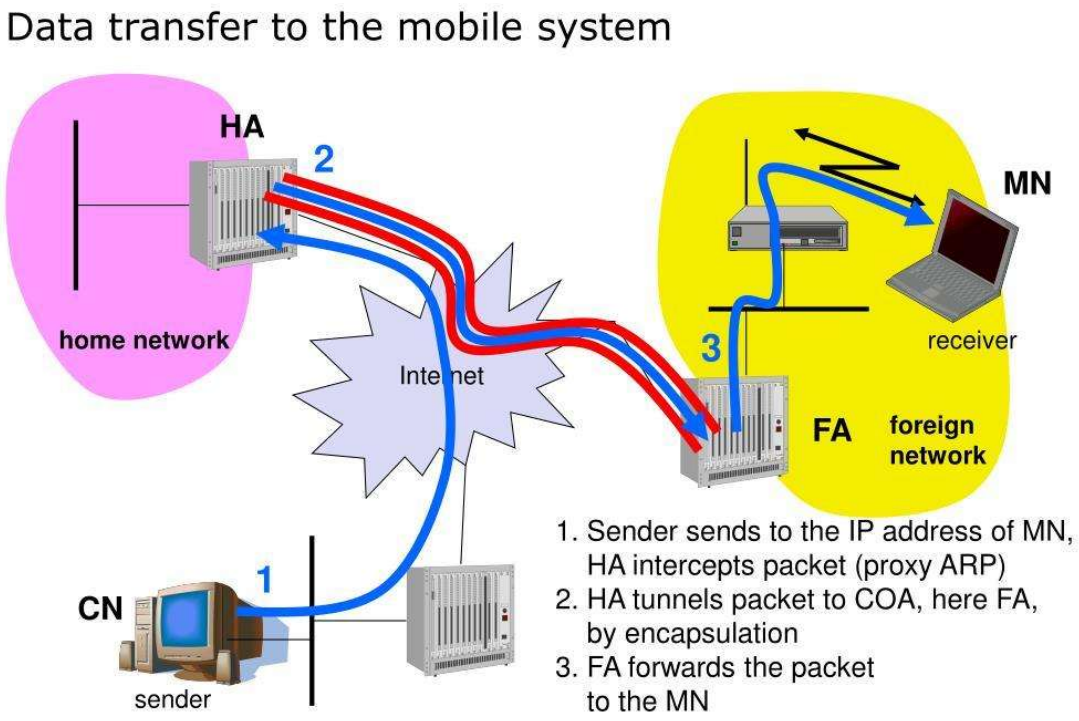
2. Mobile IP Architecture

Major Components

Component	**Function**
Mobile Node (MN)	Device that moves between networks.
Home Agent (HA)	Maintains the mobile node's current location.
Foreign Agent (FA)	Helps the mobile node in a visited network in traditional Mobile IPv4.

Correspondent Node (CN)	Communicates with the mobile node.
Care-of Address (CoA)	Temporary address representing the mobile node's current location.

Mobile IP Architecture Diagram



Explanation:

1. The Mobile Node has a permanent Home Address.
2. When it enters a foreign network, it obtains a Care-of Address.
3. The Home Agent stores the binding between the Home Address and CoA.
4. The Correspondent Node sends packets to the Home Address.
5. The Home Agent intercepts these packets.
6. The Home Agent tunnels packets to the Care-of Address.
7. The Foreign Agent forwards the packets to the Mobile Node.

3. Agent Discovery

Agent discovery is the process by which a mobile node identifies whether it is connected to its home network or a foreign network.

Two methods:

- Agent Advertisement: Agents periodically announce their availability.
- Agent Solicitation: A mobile node requests agent information when necessary.

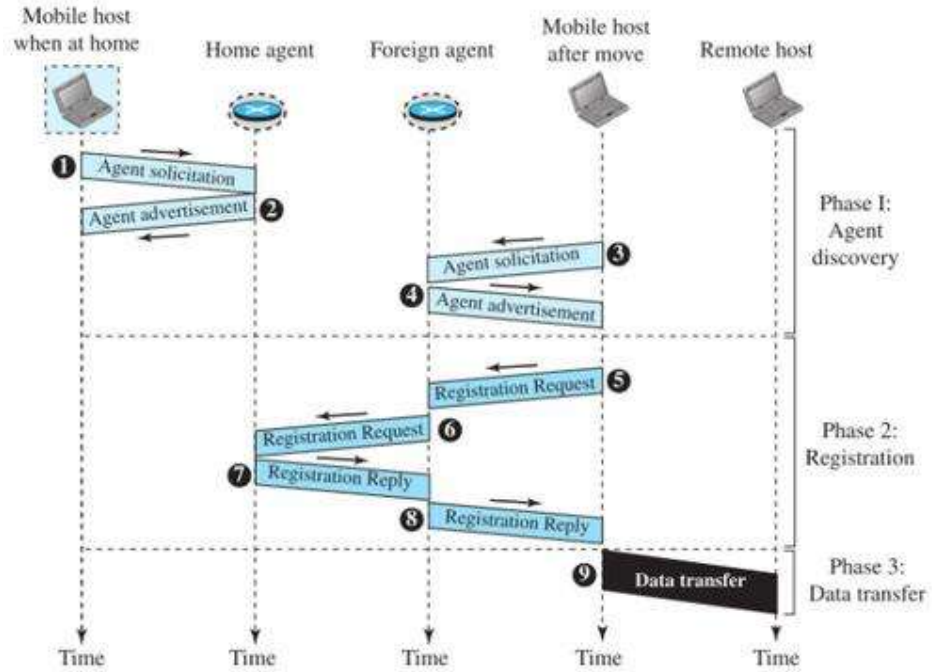
4. Mobile IP Registration

Registration updates the Home Agent with the current Care-of Address of the mobile node.

Registration steps:

1. Mobile Node enters a foreign network.
2. It obtains a Care-of Address.
3. It sends a Registration Request to the Home Agent.
4. The Home Agent authenticates the request.
5. The Home Agent updates its binding table.
6. The Home Agent sends a Registration Reply.
7. The mobile node becomes registered.

Registration Flow Diagram



5. Packet Delivery and Tunnelling

Tunnelling is the process of encapsulating an original IP packet inside another IP packet to forward it to the mobile node's current location.

Packet delivery steps:

1. Correspondent Node sends a packet to the Mobile Node's Home Address.
2. Home Agent receives the packet.
3. Home Agent checks the registered Care-of Address.
4. Original packet is encapsulated.
5. Encapsulated packet is forwarded through the tunnel.
6. Tunnel endpoint decapsulates the packet.
7. Packet is delivered to the Mobile Node.

Packet Flow

Correspondent Node

|

v

Home Agent

|

v

Encapsulation

|

v

IP Tunnel

|

v

Foreign Agent

|

v

Mobile Node

6. Pseudocode – Mobile IP Location Update

ALGORITHM MobileIPLocationUpdate(MN, HA, CurrentNetwork)

INPUT:

MN – Mobile Node

HA – Home Agent

CurrentNetwork – Current network

OUTPUT:

Registered Care-of Address and Status

BEGIN

IF CurrentNetwork = MN.HomeNetwork THEN

    CoA ← MN.HomeAddress

    RegistrationStatus ← HOME

ELSE

    CoA ← ObtainCareOfAddress(CurrentNetwork)

    SEND RegistrationRequest(MN.HomeAddress, CoA) TO HA

    WAIT FOR RegistrationReply

    IF RegistrationReply = ACCEPTED THEN

        RegistrationStatus ← REGISTERED

    ELSE

RegistrationStatus ← FAILED

END IF

END IF

RETURN CoA, RegistrationStatus

END

7. Mobile IP Challenges – Student B

1. Registration Delay

Registration takes time because the mobile node must communicate with the Home Agent.

Solution: Use efficient registration and reduce unnecessary updates.

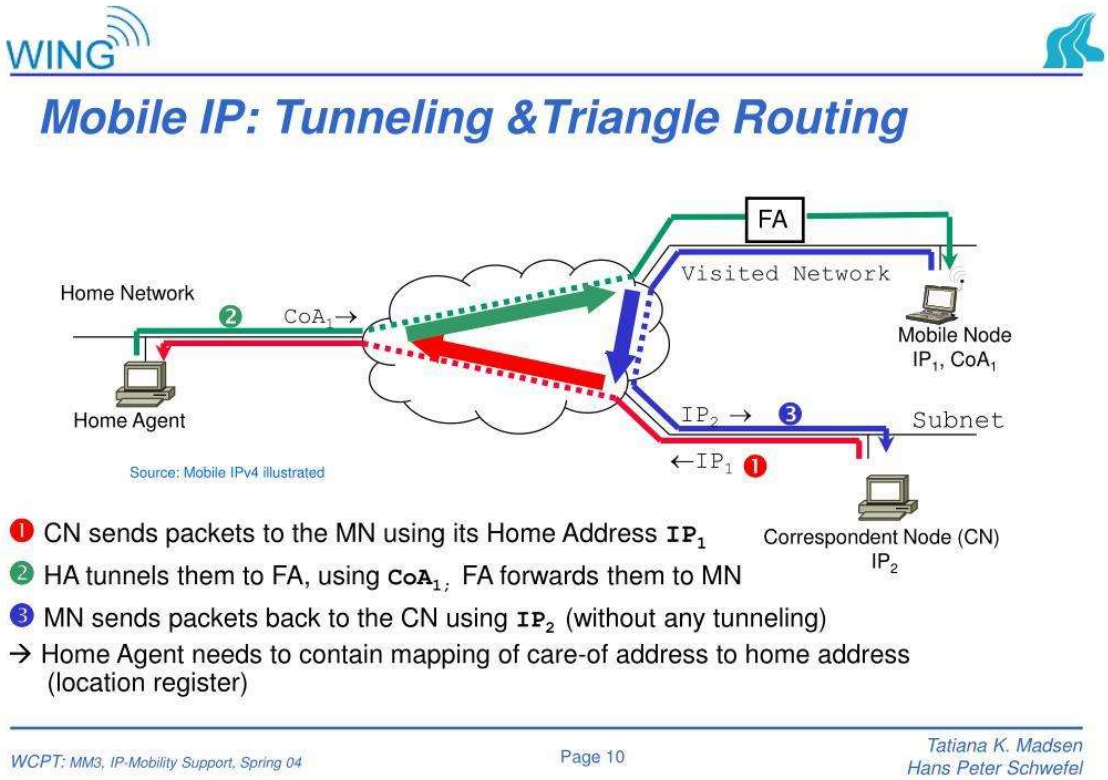
2. Tunnelling Overhead

Encapsulation adds extra IP headers, increasing packet size and network overhead.

Solution: Use efficient tunnelling techniques.

3. Triangular Routing

Packets travel from the Correspondent Node to the Home Agent and then to the Mobile Node, creating an indirect path.



Solution: Route optimization can allow the Correspondent Node to send packets more directly to the mobile node's current location, where supported.

4. Packet Loss During Mobility

Packets may be lost when the mobile node moves between networks.

Solution: Buffer packets, retransmit lost packets or use suitable forwarding mechanisms.

5. Security Issues

Unauthorized users may send fake registration messages.

Solution: Use authentication and secure mobility messages.

6. Scalability

A large number of mobile nodes can increase registration and signalling traffic.

Solution: Use efficient mobility management and reduce unnecessary registration updates.

8. Joint Design Question

Question: If a mobile user frequently changes its network of attachment during an active communication session, what problems can occur in Mobile IP and how can they be reduced?

Answer:

Frequent movement between networks can create the following problems:

1. Repeated registration requests increase signalling overhead.

2. Handoff may introduce communication delay.
3. Packets may be lost during network switching.
4. Frequent Care-of Address changes increase Home Agent updates.
5. Tunnelling creates additional network overhead.

Proposed improvements:

- Use fast handoff mechanisms to reduce switching delay.
- Buffer packets during network transitions.
- Authenticate registration messages.
- Use route optimization to reduce indirect routing.
- Reduce unnecessary registration updates.

9. Conclusion

Mobile IP provides location-independent communication by allowing mobile devices to retain their permanent IP addresses while moving between networks.

The Home Agent maintains the mobile node's location, registration updates the Care-of Address, and tunnelling forwards packets to the current network. Efficient registration, authentication, buffering and route optimization help reduce mobility-related problems.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks (7 Marks)

1. Introduction

Wireless networks are mainly classified into two types based on their communication architecture:

1. Cellular Wireless Networks
2. Ad Hoc Wireless Networks

Cellular networks use fixed infrastructure such as base stations, whereas ad hoc networks communicate directly between mobile nodes without requiring fixed infrastructure.

2. Cellular Wireless Network

A cellular network is a wireless communication system that divides a geographical area into smaller regions called cells. Each cell is controlled by a Base Station (BS).

Examples: 4G LTE, 5G and GSM networks.

Components

Component	**Function**

Mobile Station	Mobile device used for communication.
Base Station	Provides wireless connectivity to mobile devices.
Base Station Controller	Manages base station operations in applicable cellular architectures.
Mobile Switching Centre	Handles call and mobility management in traditional cellular systems.
Core Network	Connects cellular users to other networks.

Working of Cellular Networks

1. The geographical area is divided into cells.
2. Each cell contains a base station.
3. Mobile devices connect to the nearest suitable base station.
4. The base station forwards communication to the core network.
5. The core network delivers data to the destination.
6. When the user moves between cells, handoff transfers the connection to another base station.

Advantages

- Large geographical coverage.
- Centralized network management.
- Reliable communication with functioning infrastructure.
- Supports a large number of mobile users.
- Suitable for voice calls and mobile Internet.

Disadvantages

- Requires fixed infrastructure.
- Installation and maintenance costs are high.
- Base station failure can affect communication in its coverage area.
- Network performance may decrease during congestion.

3. Ad Hoc Wireless Network

An ad hoc wireless network is a temporary wireless network formed by mobile devices without requiring fixed infrastructure or centralized administration.

Each mobile node can act as both a host and a router.

Examples:

- Military communication networks.
- Disaster recovery communication.
- Temporary conference networks.
- Wireless sensor networks (in suitable ad hoc configurations).

Working of Ad Hoc Networks

1. Mobile nodes discover nearby nodes.
2. Nodes establish wireless communication links.
3. If the destination is outside direct communication range, intermediate nodes forward packets.
4. Routing protocols discover suitable paths.
5. When a node moves, the network updates or discovers another route.
6. Communication continues through available paths.

Multi-Hop Communication

Multi-hop communication means transmitting data through multiple intermediate nodes to reach a destination.

Example:

Node A → Node B → Node C → Node D

Here:

- Node A is the source.
- Node B and Node C are intermediate routers.
- Node D is the destination.

Node A cannot directly reach Node D, so packets travel through B and C.

Advantages

- No fixed infrastructure is required.
- Quick deployment.
- Supports communication in disaster areas.
- Flexible network topology.
- Intermediate nodes can forward packets.

Disadvantages

- Frequent topology changes.
- Routing becomes difficult when nodes move.
- Limited battery power.
- Wireless interference may affect communication.
- Network performance may decrease as the number of nodes increases.

4. Cellular vs Ad Hoc Wireless Networks – Comparison

Parameter	**Cellular Network**	**Ad Hoc Network**
Infrastructure	Requires base stations and fixed infrastructure.	No fixed infrastructure required.
Communication	Through base stations and core network.	Direct or multi-hop node communication.
Routing	Infrastructure-assisted routing.	Distributed routing.
Mobility	Managed through cellular mobility mechanisms.	Managed through route discovery and maintenance.
Topology	Planned cell-based topology.	Dynamic topology.
Scalability	Supports large-scale deployment with infrastructure.	Depends on node density and routing capacity.
Handoff	Between base stations or cells.	Routes and communication links change as nodes move.
Failure Handling	Infrastructure failure may affect service.	Alternative peer routes may be available.
Applications	Mobile Internet, voice calls and 5G.	Disaster recovery, military and temporary networks.

5. Technical Diagram – Communication Difference

Cellular Network

Mobile device communicates through a base station.

Ad Hoc Network

Mobile nodes communicate directly or through intermediate nodes.

6. Student A – Cellular Network Investigation

Role of Base Stations

A base station provides wireless communication between mobile devices and the network infrastructure.

Its main functions are:

- Establishing wireless connections.
- Receiving and transmitting data.
- Managing radio resources.
- Supporting handoff.
- Connecting mobile users to the core network.

Centralized Communication

In cellular networks, communication is controlled and supported by centralized network infrastructure.

Mobile devices communicate through base stations rather than depending entirely on peer-to-peer routing.

Handoff Between Cells

Handoff occurs when a mobile device moves from one cell to another.

Steps:

1. Mobile device connects to BS1.
2. BS1 signal becomes weak.
3. Device detects BS2.
4. Network selects BS2.
5. Connection transfers to BS2.
6. Communication continues.

Scalability and Coverage

Cellular networks provide wide geographical coverage by deploying multiple interconnected cells. Additional base stations can increase coverage and capacity.

Suitable Applications

- Mobile communication.
- Video streaming.
- Online banking.

- GPS navigation.
- Mobile Internet access.

7. Student B – Ad Hoc Network Investigation

Communication Between Mobile Nodes

Mobile nodes communicate directly when they are within transmission range. Otherwise, intermediate nodes forward packets.

Dynamic Topology

Ad hoc network topology changes frequently because mobile devices move.

When a node moves away:

- Existing links may break.
- Routes may become invalid.
- Routing protocols discover alternative paths.

Multi-Hop Communication

Intermediate nodes forward packets from source to destination when direct communication is unavailable.

Absence of Fixed Infrastructure

Ad hoc networks do not require permanent base stations or centralized controllers. Nodes organize and manage communication cooperatively.

Suitable Applications

- Disaster recovery.
- Military communication.
- Temporary field networks.
- Search and rescue operations.
- Emergency communication.

8. Joint Engineering Decision

Question: For an emergency/disaster environment where fixed infrastructure may be unavailable, which network architecture would you select and why?

Answer:

I would select an Ad Hoc Wireless Network (MANET) because it does not depend on fixed infrastructure such as base stations or centralized network controllers.

During natural disasters, cellular towers and communication infrastructure may become damaged. Ad hoc networks allow rescue teams, ambulances, police vehicles and medical personnel to communicate directly or through intermediate mobile nodes.

Reasons for selecting Ad Hoc Networks:

1. No fixed infrastructure is required.
2. Can be deployed quickly.
3. Supports multi-hop communication.
4. Mobile nodes can act as routers.
5. Alternative routes may be available when a link fails.
6. Suitable for emergency and disaster recovery situations.

Example:

Ambulance → Rescue Team → Police Vehicle

|

v

Medical Team

The nodes forward information through available wireless links even when cellular infrastructure is unavailable.

9. Pseudocode – Selecting Wireless Architecture

ALGORITHM SelectWirelessArchitecture(Scenario)

INPUT:

InfrastructureAvailability

NodeMobility

DeploymentTime

CoverageRequirement

MultiHopRequirement

OUTPUT:

Recommended Architecture

```
BEGIN

IF InfrastructureAvailability = AVAILABLE AND
   CoverageRequirement = LARGE THEN

    Architecture ← CELLULAR

ELSE IF DeploymentTime = IMMEDIATE AND
        InfrastructureAvailability = UNAVAILABLE THEN

    Architecture ← AD_HOC

ELSE IF MultiHopRequirement = REQUIRED THEN

    Architecture ← AD_HOC

ELSE

    Architecture ← CELLULAR

END IF

RETURN Architecture

END
```

10. Conclusion

Cellular networks provide wide coverage, centralized management and reliable communication through fixed infrastructure. Ad hoc networks provide flexible, infrastructure-less communication using direct and multi-hop connections between mobile nodes.

Therefore, cellular networks are suitable for regular mobile communication, whereas ad hoc networks are more suitable for emergency situations where fixed infrastructure is unavailable.

Q4 – Applications of Mobile IP and Routing Requirements (7 Marks)

1. Introduction

Mobile IP is used in applications where mobile devices need continuous network communication while moving from one location to another.

It allows a mobile node to maintain its permanent IP address even when it changes its network attachment.

Real-world applications:

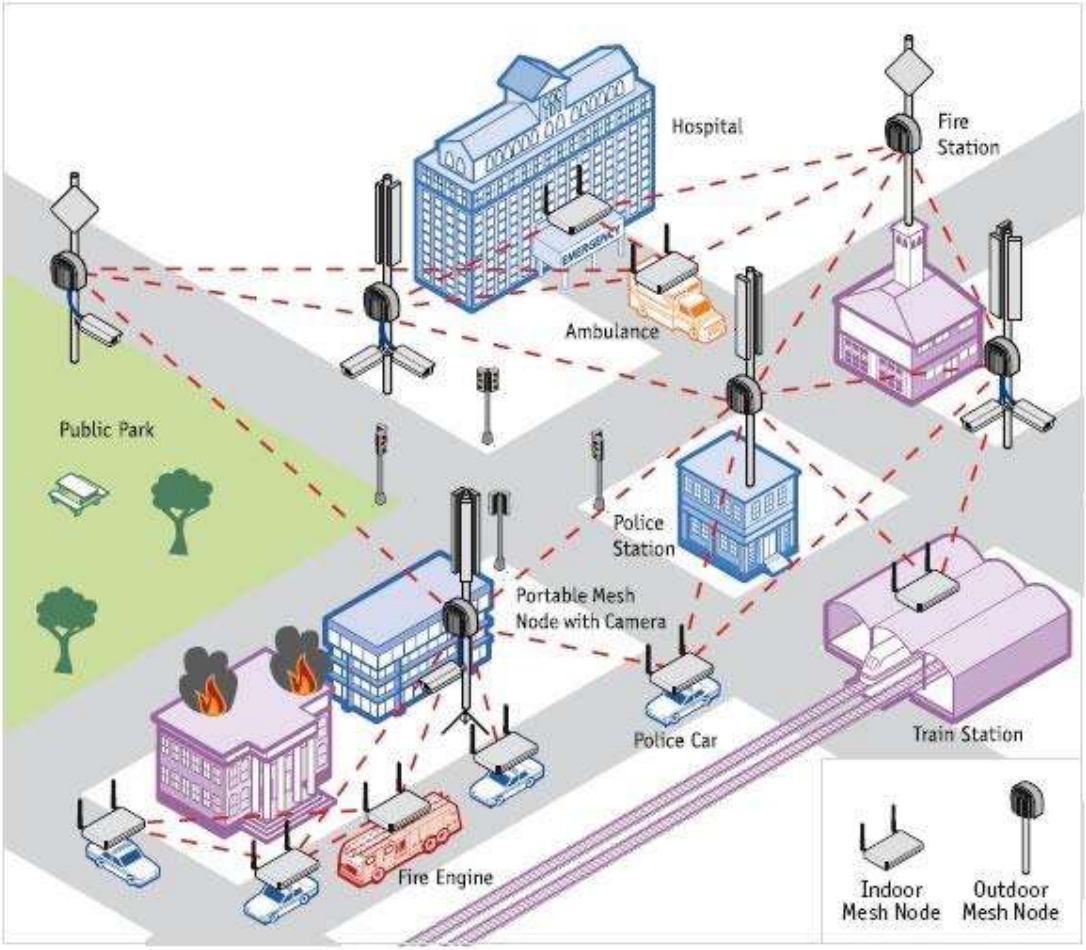
1. Emergency communication.
2. Mobile healthcare.
3. Intelligent transportation systems.
4. Mobile Internet access.
5. Vehicle and fleet monitoring.
6. Remote field engineering.

2. Selected Application – Emergency Communication System

An emergency communication system connects ambulances, police vehicles, rescue teams and medical personnel during emergency situations.

Mobile IP helps maintain communication when emergency vehicles move between different network areas.

3. Mobile IP Application Architecture – Student B



Network Components:

Component	**Role**
Mobile Node	Ambulance, police vehicle or rescue team device.
Home Agent	Maintains the permanent location information of mobile nodes.
Foreign Agent	Supports mobile nodes in a visited network in traditional Mobile IPv4.
Correspondent Node	Hospital server or emergency control centre.
Care-of Address	Represents the current network location of a mobile node.
Wireless Network	Provides communication between mobile devices and network infrastructure.

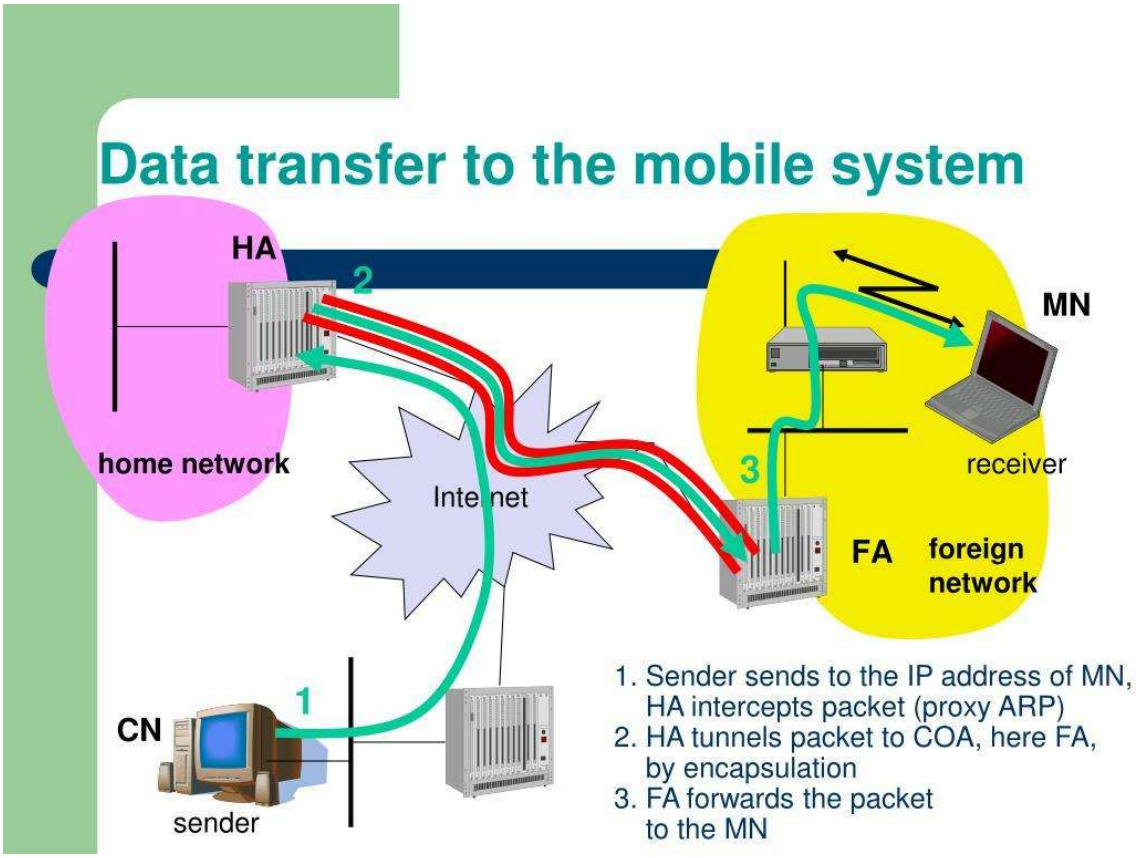
4. How Mobile IP Supports User Mobility

Mobile IP supports communication when an ambulance moves from one network to another.

Working process:

1. Ambulance connects to its home network.
2. Ambulance moves into another network.
3. It obtains a new Care-of Address.
4. It sends a registration request to its Home Agent.
5. Home Agent updates the mobile node's location.
6. Emergency data packets are forwarded through tunnelling.
7. The hospital server continues receiving information.

5. Communication Flow



Example:

An ambulance travels from Area A to Area B while transmitting a patient's emergency information to a hospital.

- Area A uses BS1.
- Area B uses BS2.
- When the ambulance moves to Area B, it obtains a new Care-of Address.

- The Home Agent updates the location.
- The hospital continues receiving emergency information.

6. Performance and Requirement Analysis – Student A

A. Communication Delay

Communication delay is the time required for data packets to travel from the sender to the receiver.

During mobility, delay may increase because of:

- Network discovery.
- Care-of Address acquisition.
- Registration with Home Agent.
- Packet tunnelling.
- Handoff between networks.

Solution: Use fast handoff mechanisms and efficient registration.

B. Packet Delivery and Connectivity

Packet delivery means successfully transmitting data packets to the destination.

During movement, packets may be lost due to temporary disconnection.

Solution: Buffer packets and retransmit lost packets when appropriate.

C. Mobility Requirements

- Continuous connectivity.
- Fast handoff.
- Automatic location updates.
- Stable IP communication.
- Efficient packet forwarding.

D. Security Requirements

Emergency communication contains important patient and rescue information.

Required security mechanisms:

1. Authentication of mobile devices.
2. Encryption of sensitive information.
3. Secure registration.

4. Protection against unauthorized access.

E. Reliability Requirements

- Minimum packet loss.
- High network availability.
- Backup communication paths.
- Reliable data transmission.
- Quick recovery from network failure.

7. Pseudocode – Mobile IP Emergency Communication

```
ALGORITHM ManageEmergencyCommunication(MN, HA, CN)

INPUT:
MN – Emergency Mobile Node
HA – Home Agent
CN – Hospital or Emergency Server

OUTPUT:
Communication status

BEGIN

INITIALISE RegistrationStatus ← NOT_REGISTERED
INITIALISE CommunicationStatus ← DISCONNECTED

DETECT CurrentNetwork

IF CurrentNetwork ≠ MN.HomeNetwork THEN

    CoA ← ObtainCareOfAddress(CurrentNetwork)
```

```
    SEND RegistrationRequest(MN.HomeAddress, CoA) TO HA

    WAIT FOR RegistrationReply

    IF RegistrationReply = ACCEPTED THEN

        RegistrationStatus ← REGISTERED

    ELSE

        RETURN REGISTRATION_FAILURE

    END IF

ELSE

    RegistrationStatus ← HOME

END IF

IF RegistrationStatus = REGISTERED OR
   RegistrationStatus = HOME THEN

    SEND EmergencyData TO CN

    IF PacketDeliverySuccessful THEN

        CommunicationStatus ← CONNECTED
```

```
    ELSE

        BUFFER EmergencyData
        RETRY transmission when connectivity is restored

    END IF

END IF

RETURN CommunicationStatus

END
```

8. Evaluation of Mobile IP Suitability

Requirement	**Mobile IP Support**
Mobility	Maintains permanent Home Address.
Connectivity	Supports communication across networks.
Packet Delivery	Uses tunnelling to forward packets.
Location Update	Registration updates current location.
Security	Authentication protects registration.
Reliability	Buffering and retransmission can reduce packet loss.

Limitation: Mobile IP depends on network infrastructure and does not by itself guarantee uninterrupted communication during complete network failure.

9. Conclusion

Mobile IP is suitable for emergency communication because it supports mobile devices moving between different networks while maintaining their permanent IP addresses.

It helps ambulances, rescue teams and hospitals exchange critical information. Fast handoff, secure registration, efficient routing and packet buffering can improve communication performance.

Q5 – Proactive Routing Protocols in Mobile Networks (6 Marks)

1. Introduction to Proactive Routing

Proactive routing is a routing approach in which every node continuously maintains routing information about available destinations before data transmission is requested.

It uses periodic routing updates to maintain routing tables.

Example: Destination-Sequenced Distance Vector (DSDV).

Characteristics

1. Maintains routing tables continuously.
2. Periodically exchanges routing information.
3. Provides routes before data transmission.
4. Reduces initial route discovery delay.
5. Generates continuous control-message overhead.
6. Uses sequence numbers to maintain fresh routes.

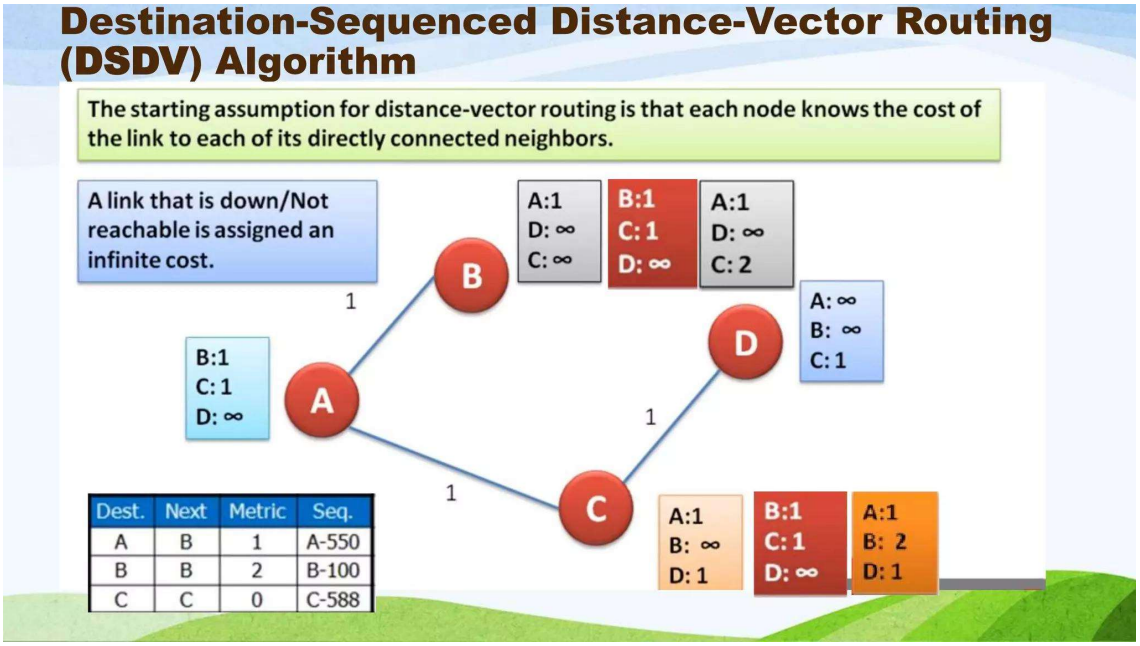
2. DSDV Routing Protocol

DSDV stands for Destination-Sequenced Distance Vector.

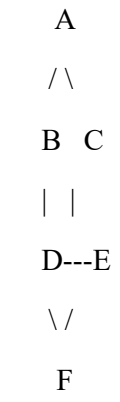
It is a proactive routing protocol used in mobile ad hoc networks.

DSDV maintains a routing table at each node containing information about destinations, next hops, hop counts and sequence numbers.

3. DSDV Network Architecture



Consider the following network:



Each node maintains routing information about other reachable nodes.

For example, Node A can reach Node F through intermediate nodes.

4. DSDV Routing Table

Each routing table contains:

Parameter	Description
Destination	Target node.
Next Hop	Next node used to forward packets.
Hop Count	Number of intermediate links.
Sequence Number	Indicates freshness of route information.
Route State	Valid or invalid route status.

5. Role of Sequence Numbers

Sequence numbers help identify the freshness of routing information.

- A higher sequence number indicates newer route information.
- If sequence numbers are equal, the route with a smaller hop count is preferred.
- Older routing information is rejected.
- Sequence numbers help prevent routing loops.

Example:

Node A receives two routes to Node D.

Route	Sequence Number	Hop Count
Route 1	10	3

Route 2	12	4

Node A selects Route 2 because it has a newer sequence number, even though it has more hops.

6. DSDV Routing Table Maintenance

Student A – Protocol Design

DSDV maintains routing tables through periodic updates.

Steps:

1. Each node initializes its routing table.
2. Nodes broadcast routing information.
3. Neighbouring nodes receive updates.
4. Nodes compare sequence numbers.
5. Newer routing information replaces older information.
6. If sequence numbers are equal, the route with fewer hops is selected.
7. Expired or invalid routes are removed.

7. Pseudocode – DSDV Route Update

ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)

INPUT:

Node – Receiving node

ReceivedRoute – Destination, NextHop,

HopCount, SequenceNumber

OUTPUT:

Updated routing table

BEGIN

Destination ← ReceivedRoute.Destination

```
NewHopCount ← ReceivedRoute.HopCount + 1

NewSequence ← ReceivedRoute.SequenceNumber

IF Destination NOT IN RouteTable THEN

   ADD ReceivedRoute TO RouteTable

ELSE IF NewSequence >
      RouteTable[Destination].SequenceNumber THEN

   REPLACE RouteTable[Destination]

ELSE IF NewSequence =
      RouteTable[Destination].SequenceNumber AND
      NewHopCount < RouteTable[Destination].HopCount THEN

   REPLACE RouteTable[Destination]

END IF

RETURN RouteTable

END
```

8. Routing Information Before and After Topology Change

Consider this network:

Before topology change:

A → B → C → D

Node A reaches Node D through B and C.

Destination	Next Hop	Hop Count	Sequence Number
B	B	1	4
C	B	2	6
D	B	3	8

After topology change:

Suppose the link between B and C fails, but another path is available:

A → E → D

Node A receives updated routing information.

Destination	Next Hop	Hop Count	Sequence Number
E	E	1	10
D	E	2	12
C	-	-	Invalid

The routing table is updated to use the available path through E.

Illustrative topology and routing-table values for manual analysis.

9. Student B – Performance Investigation

A. Routing Control Overhead

Routing control overhead refers to the additional network traffic generated by routing updates.

DSDV periodically broadcasts routing information even when there is no data transmission.

Effect: Consumes bandwidth and battery power.

B. Route Availability

DSDV maintains routes before communication begins.

Therefore, a valid route may already be available when a source wants to send data.

C. Convergence Behaviour

Convergence is the process of updating routing tables after a network topology change.

When a link fails, nodes exchange routing updates until consistent route information is established.

D. Control Message Transmission

DSDV uses periodic and triggered routing updates.

- Periodic updates maintain routing information.
- Triggered updates inform nodes about significant topology changes.

E. Different Network Sizes

Network Size	DSDV Behaviour
Small Network	Low routing overhead.
Medium Network	Moderate routing overhead.
Large Network	More update messages and bandwidth usage.

F. Delay and Packet Delivery

Advantages:

- Low initial route discovery delay.
- Routes are already available.
- Suitable for frequent communication.

Disadvantages:

- Periodic updates consume bandwidth.
- Rapid topology changes may cause stale routes.
- Packet delivery may be affected during convergence.

10. Proactive Routing Performance Summary

Parameter	DSDV Behaviour
Route Discovery	Not required before every transmission.
Initial Delay	Low when a valid route exists.
Control Overhead	High due to periodic updates.
Mobility Handling	Updates routing tables after topology changes.
Bandwidth	Consumes bandwidth for control messages.
Suitable Network	Small or moderately dynamic networks with frequent communication.

11. Conclusion

DSDV is a proactive routing protocol that maintains routing tables continuously using periodic updates and sequence numbers.

It provides routes before data transmission and reduces initial communication delay. However, it creates control-message overhead, especially in large and highly dynamic networks.

Therefore, DSDV is suitable when communication is frequent and low route-discovery delay is important.

Q6 – Reactive Routing and Protocol Comparison (7 Marks)

1. Introduction to Reactive Routing

Reactive routing is a routing approach in which routes are discovered only when a source node wants to communicate with a destination.

Unlike proactive routing, reactive routing does not continuously maintain routes to every destination.

It is also called On-Demand Routing.

Example: Ad Hoc On-Demand Distance Vector (AODV).

Characteristics

1. Routes are discovered only when required.
2. Uses Route Request and Route Reply messages.
3. Reduces unnecessary routing updates.
4. Maintains active routes.
5. Supports dynamic mobile networks.
6. May introduce initial route discovery delay.

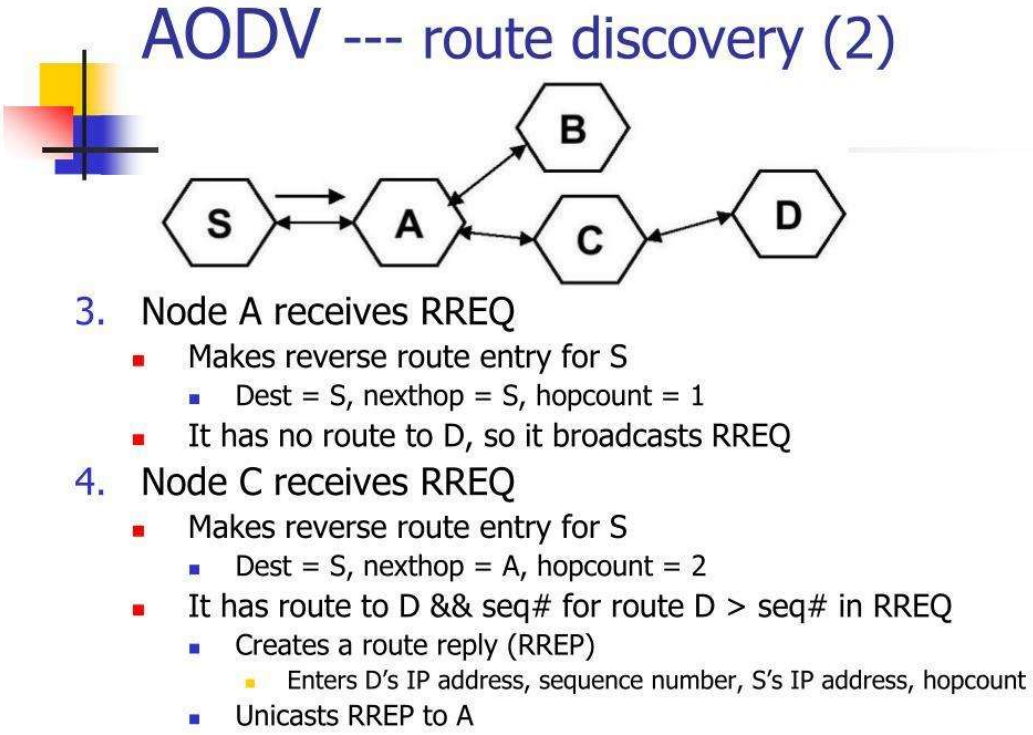
2. AODV Routing Protocol

AODV stands for Ad Hoc On-Demand Distance Vector.

It is a reactive routing protocol used in mobile ad hoc networks.

AODV discovers routes when a source node needs to send data to a destination.

3. AODV Network Architecture



Consider the following network:

```
   A
  / \
 S   B
  \ / \
   C   D
```

Here:

- S = Source node.
- D = Destination node.
- A, B and C = Intermediate nodes.

The source node discovers a route to the destination using AODV.

4. Important AODV Messages

A. Route Request (RREQ)

RREQ is a message sent by the source node to discover a route to a destination.

Working:

1. Source checks its routing table.
2. If no valid route exists, it generates RREQ.
3. RREQ is broadcast to neighbouring nodes.
4. Intermediate nodes record reverse-route information.
5. RREQ reaches the destination or a node with a valid route.

B. Route Reply (RREP)

RREP is a message sent by the destination or an intermediate node with a valid route.

Working:

1. Destination receives RREQ.
2. Destination generates RREP.
3. RREP travels back through the reverse path.
4. Intermediate nodes update their routing tables.
5. Source receives RREP.
6. A valid route is established.

C. Route Error (RERR)

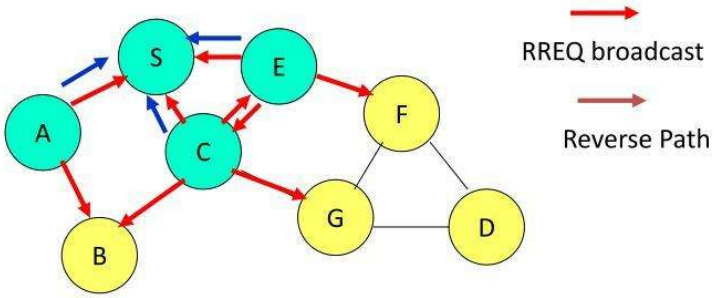
RERR is a message used to report a broken or unavailable route.

Working:

1. A node detects link failure.
2. The affected route is marked invalid.
3. RERR is generated.
4. RERR is sent to affected source nodes.
5. Source performs route discovery again if communication is still required.

5. AODV Route Discovery Process

AODV Route Discovery



- Uses hop-by-hop routing.
- Reverse paths are formed when a node hears a route request.
- Each node forwards the request only once (pure flooding).

Example:

Source S wants to send data to Destination D.

Step 1: S checks its routing table.

Step 2: No valid route exists.

Step 3: S broadcasts RREQ to neighbouring nodes.

Step 4: Intermediate nodes forward RREQ.

Step 5: D receives RREQ and generates RREP.

Step 6: RREP travels back to S.

Step 7: S establishes the route and starts transmitting data.

Final route:

S → B → D

6. Pseudocode – AODV Route Discovery

ALGORITHM ReactiveRouteDiscovery(Source, Destination)

INPUT:

Source – Requesting node

Destination – Target node

OUTPUT:

Established route or ROUTE_NOT_FOUND

BEGIN

IF RouteTable[Destination].State = VALID THEN

    RETURN RouteTable[Destination]

END IF

RouteRequestID ← RouteRequestID + 1

Create RouteRequest(Source, Destination, RouteRequestID)

BROADCAST RouteRequest TO Neighbors

START TIMER

WAIT FOR RouteReply OR RouteFailure

IF RouteReply RECEIVED THEN

    Establish reverse path

    Establish forward path

RouteTable[Destination].State ← VALID

RETURN RouteTable[Destination]

ELSE

RETURN ROUTE_NOT_FOUND

END IF

END

7. AODV Route Maintenance

Route maintenance ensures that active routes remain available during communication.

Steps:

1. Monitor the active route.
2. Detect link failures.
3. Mark the failed route as invalid.
4. Check whether an alternative route exists.
5. If available, select the alternative route.
6. Otherwise, generate RERR.
7. Initiate new route discovery if communication is still required.

Pseudocode – Route Maintenance

ALGORITHM MaintainReactiveRoute(Source, Destination)

INPUT:

Source and Destination

OUTPUT:

```
Route status

BEGIN

MONITOR active route

IF LinkFailureDetected THEN

    Mark failed link as INVALID

    IF AlternateValidRoute EXISTS THEN

        Select alternate route

        RETURN ALTERNATE_ROUTE_USED

    ELSE

        Generate RouteError

        SEND RouteError TO affected source

        IF Source still requires communication THEN

            Initiate new RouteDiscovery

        ELSE

            RETURN ROUTE_TERMINATED
```

END IF

END IF

END IF

RETURN ROUTE_ACTIVE

END

8. Effect of Mobility on Route Discovery

Mobility affects AODV because mobile nodes frequently change their locations.

Effects:

- Existing wireless links may break.
- Routes may become invalid.
- New route discovery may be required.
- RREQ and RREP messages may increase.
- Communication delay may increase.
- Packet loss may occur during route repair.

Solution: Use route maintenance, alternative paths and efficient route discovery.

9. Student B – DSDV and AODV Comparison

Parameter	**DSDV (Proactive)**	**AODV (Reactive)**
Routing Type	Proactive	Reactive
Route Discovery	Routes maintained continuously.	Routes discovered when required.
Routing Table	Maintains routes to destinations.	Maintains routes for active communication.
Control Overhead	Periodic updates generate overhead.	Control messages generated on demand.

Initial Delay	Usually low if route is valid.	Higher when route discovery is required.
Mobility Handling	Routing tables are updated.	Broken routes trigger repair or rediscovery.
Bandwidth Usage	Periodic bandwidth consumption.	Generally lower when traffic is intermittent.
Route Availability	Routes available in advance.	Routes available after discovery.
Suitable Applications	Frequent communication.	Intermittent communication and dynamic networks.

10. Performance Analysis

A. Route Discovery Mechanism

- DSDV maintains routes before communication.
- AODV discovers routes only when communication is required.

B. Routing Table Requirements

- DSDV continuously maintains destination routing information.
- AODV maintains information needed for active routes and route discovery.

C. Control Overhead

DSDV generates periodic routing updates even when there is no data transmission.

AODV generates RREQ, RREP and RERR messages when needed.

Therefore, AODV can reduce unnecessary control traffic when communication is infrequent.

D. Initial Communication Delay

- DSDV generally has lower initial delay because routes are already available.
- AODV introduces route discovery delay when no valid route exists.

E. Mobility Handling

- DSDV updates routing tables when topology changes.
- AODV detects broken active links and initiates route repair or rediscovery.

F. Application Scenarios

Network Condition	**Suitable Protocol**
Frequent data communication	DSDV

Infrequent data communication	AODV
Limited bandwidth	AODV in intermittent traffic conditions
Low initial delay required	DSDV
Dynamic network with intermittent traffic	AODV

11. Joint Question – Protocol Selection

Question: For a highly dynamic mobile network with frequent topology changes, would you select DSDV or AODV? Justify your decision using routing overhead, delay and mobility considerations.

Answer:

I would select AODV (Reactive Routing Protocol) for a highly dynamic mobile network, particularly when communication is intermittent.

Reasons:

1. Routing Overhead: AODV does not continuously broadcast routing-table updates, reducing unnecessary control traffic.
2. Delay: AODV introduces route discovery delay when a route is unavailable, but avoids maintaining unused routes.
3. Mobility: When a link breaks, AODV detects the failure and initiates route repair or new route discovery.
4. Bandwidth: It can conserve bandwidth when communication is not continuous.
5. Route Maintenance: RERR messages help inform source nodes about broken routes.

Limitation: In a highly mobile network with continuous high-volume communication, frequent route failures can cause repeated RREQ broadcasts and increased delay. DSDV may be preferable when low initial transmission delay is more important and updates can converge quickly.

12. Conclusion

AODV is a reactive routing protocol that discovers routes only when required. It uses RREQ, RREP and RERR messages for route discovery and maintenance.

DSDV maintains routes continuously, whereas AODV discovers routes on demand. AODV is suitable for intermittent communication in dynamic mobile networks because it can reduce unnecessary routing overhead.

Q7 – Design a Mobile Emergency Communication Network (10 Marks)

Scenario

During a natural disaster, ambulances, police vehicles, rescue teams and medical personnel move continuously across different areas. Fixed network infrastructure may be damaged or unavailable. Therefore, emergency teams need a reliable wireless communication network to exchange critical information with minimum delay.

Stage 1 – Independent Design Proposal

Student A – Wireless Architecture and Mobility Management

1. Proposed wireless architecture

I propose a hybrid wireless network that combines a Mobile Ad Hoc Network (MANET) with Mobile IP.

- Mobile nodes include ambulances, police vehicles, rescue teams and medical personnel.
- Wireless links allow nearby mobile nodes to communicate directly.
- A mobile gateway connects the emergency network to an available external network.
- Mobile IP maintains communication when a mobile node changes its network location.

2. Mobility management strategy

Mobile IP can be used to maintain the communication session when a mobile node moves from one network to another.

The main components are:

Component	**Function**
Mobile Node (MN)	Ambulance, police vehicle or rescue team device
Home Agent (HA)	Maintains the mobile node's permanent home address
Foreign Agent (FA)	Supports the mobile node in a visited network
Correspondent Node (CN)	Communicates with the mobile node

3. Communication model

1. The mobile node enters a new network.
2. It discovers the available agent.
3. It obtains a Care-of Address (CoA).
4. It registers the CoA with its Home Agent.

5. The Home Agent forwards packets through tunnelling.
6. The mobile node continues communication.

4. Role of Mobile IP

- Maintains communication during network movement.
- Supports continuous access to emergency information.
- Helps maintain the mobile node's permanent IP identity.
- Reduces communication interruption during handoff.

Student B – Routing Strategy and Performance

1. Proposed routing protocol: AODV

I propose using Ad Hoc On-Demand Distance Vector (AODV) routing because emergency vehicles and rescue teams move frequently.

AODV discovers routes only when communication is required.

2. Routing process

- RREQ: Route Request is broadcast to discover a route.
- RREP: Route Reply is sent when a route is found.
- RERR: Route Error is generated when a route becomes unavailable.
- Route maintenance: AODV discovers an alternative route when the current route fails.

3. Failure handling

- If a node moves away, AODV searches for another route.
- If a wireless link fails, the broken route is removed.
- If a network becomes disconnected, packets are stored temporarily where possible and forwarded when connectivity returns.
- If no alternative route exists, the source retries route discovery.

4. Performance parameters

- End-to-end delay
- Packet delivery ratio
- Routing overhead
- Route discovery time
- Network connectivity

Stage 2 – Compare the Two Designs

Comparison	Student A	Student B
Main focus	Wireless architecture and mobility	Routing and route maintenance
Architecture	Hybrid wireless architecture	MANET
Mobility management	Mobile IP	AODV route discovery
Routing	Not the primary focus	Reactive routing
Main advantage	Maintains IP communication across networks	Finds routes when required
Limitation	Requires reachable IP infrastructure and agent support	Route discovery may increase delay

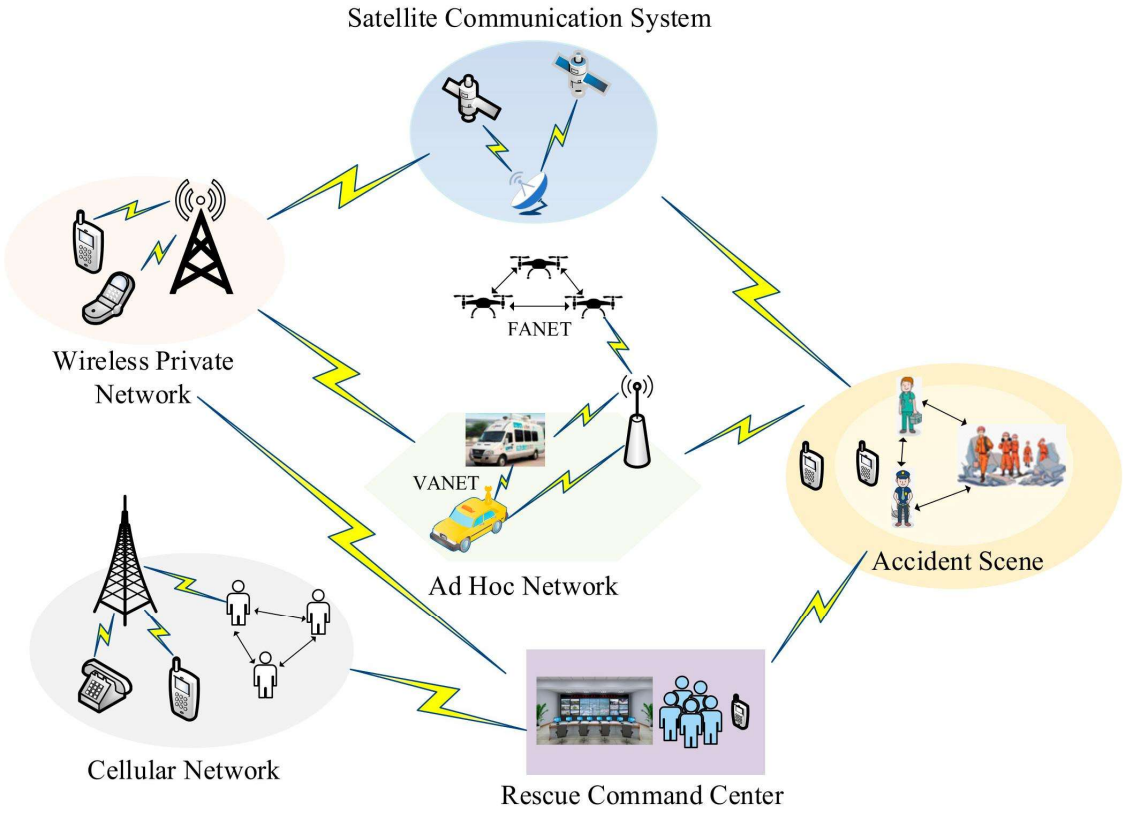
Agreement: Both designs aim to maintain communication between emergency teams while they move.

Differences: Student A focuses on maintaining communication across IP networks, whereas Student B focuses on finding routes between mobile nodes.

Joint decision: Combine both approaches. Use AODV for local emergency communication and Mobile IP when a connection to an external IP network is available.

Stage 3 – Joint Final Design

1. Final network architecture



The final architecture contains:

- Mobile emergency nodes.
- Wireless communication links.
- AODV routing between mobile nodes.
- A mobile gateway.
- Mobile IP components for external communication.

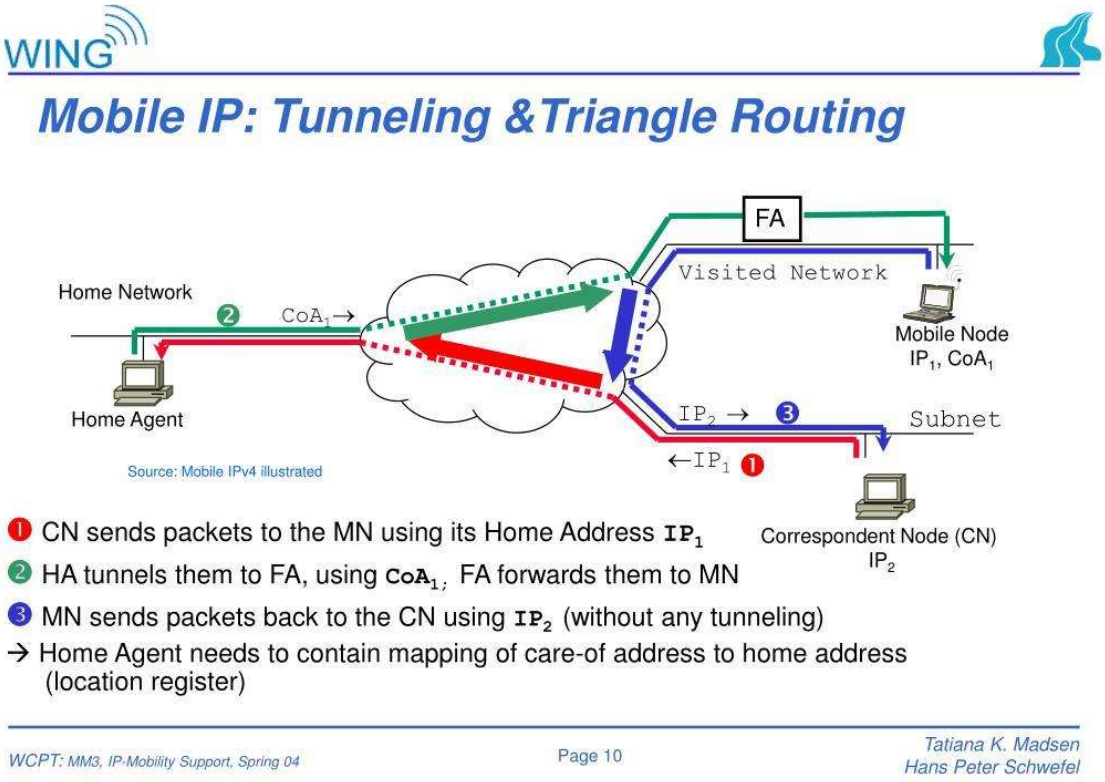
Important design consideration: Mobile IP alone cannot provide communication when all supporting IP infrastructure is unavailable. AODV allows nearby emergency nodes to communicate directly without depending on a fixed base station. Mobile IP is used when a suitable gateway and external IP connection are available.

2. Mobile IP components

Component	Role in the emergency network
MN	Ambulance or rescue team device
HA	Maintains the mobile node's home address
FA	Supports registration in a visited network

CN Hospital or emergency control centre

3. Registration and tunnelling process



1. The Mobile Node enters a foreign network.
2. It discovers the Foreign Agent.
3. The Mobile Node obtains a Care-of Address.
4. It sends a registration request to the Home Agent.
5. The Home Agent updates the mobile node's location.
6. The Correspondent Node sends data to the Home Address.
7. The Home Agent tunnels packets to the Care-of Address.
8. The Mobile Node receives the packets.

4. Selected routing protocol – AODV

AODV is selected because:

- Emergency vehicles move frequently.
- Routes may change because of node mobility.
- Routes are discovered only when required.
- Broken routes can be detected and replaced.

- It avoids maintaining unnecessary routes to every destination.

5. Failure handling

Failure	**Solution**
Node failure	Discover an alternative route
Link failure	Generate RERR and remove the broken route
Network disconnection	Buffer important packets where possible
Route failure	Start a new route discovery process
Gateway failure	Use another available gateway
Mobile node changes network	Perform Mobile IP registration when external connectivity is available

6. Performance evaluation

The following parameters are used to evaluate the network:

End-to-End Delay

$$Delay = \frac{\text{Total packet delivery time}}{\text{Number of packets received}}$$

Packet Delivery Ratio (PDR)

$$PDR = \frac{\text{Packets received}}{\text{Packets sent}} \times 100$$

Routing Overhead

$$Routing\ Overhead = \frac{\text{Routing control packets}}{\text{Total packets delivered}}$$

A successful emergency network should provide low delay, high packet delivery ratio and controlled routing overhead.

Stage 4 – Validation Responsibilities

Student A – Analytical Validation

Student A evaluates Mobile IP performance.

Parameter	**Validation**
Mobile IP overhead	Analyze additional registration and tunnelling messages
Mobility delay	Measure interruption during handoff

Packet delivery	Check whether packets reach the mobile node
Communication interruption	Analyze packet loss during movement
Mobility management	Justify how Mobile IP maintains the mobile node's identity

Student B – Experimental Validation

Student B tests the routing protocol under different network conditions.

Test cases:

Test case	**Network condition**
Test 1	10 mobile nodes
Test 2	20 mobile nodes
Test 3	30 mobile nodes
Test 4	Low mobility
Test 5	High mobility
Test 6	Link failure
Test 7	Node failure

Routing comparison:

Parameter	**DSDV (Proactive)**	**AODV (Reactive)**
Route discovery	Maintains routes periodically	Discovers routes when needed
Initial communication	Routes may already be available	Route discovery may cause delay
Control overhead	Periodic updates	Control messages during route discovery and maintenance
Mobility handling	Updates routing tables after topology changes	Repairs or rediscovers broken routes
Suitable situation	More stable communication patterns	Dynamic networks with changing routes

Experimental result table template

Actual values must be collected during simulation or testing.

Test case	Number of nodes	Average delay	PDR (%)	Routing overhead
Test 1	10	To be measured	To be measured	To be measured
Test 2	20	To be measured	To be measured	To be measured
Test 3	30	To be measured	To be measured	To be measured

Stage 5 – Cross-Defence

Question 1 – Student A: Why is AODV suitable for an emergency communication network?

Answer: AODV is suitable because emergency vehicles and rescue teams move continuously. It discovers routes when communication is required and can find alternative routes when existing links fail. This helps maintain communication in a dynamic wireless network.

Question 2 – Student B: How does Mobile IP maintain communication when a mobile node changes its network location?

Answer: Mobile IP allows a mobile node to maintain its permanent Home Address while moving between networks. The mobile node obtains a Care-of Address and registers it with the Home Agent. The Home Agent uses tunnelling to forward packets to the mobile node's current location.

Stage 6 – Joint Evaluation

Evaluation factor	Final design decision
Architecture suitability	Hybrid wireless architecture
Mobile IP requirement	Required for external IP mobility when supporting infrastructure is available
Routing selection	AODV
Communication overhead	Controlled by using on-demand route discovery
Mobility effects	Managed through AODV route maintenance and Mobile IP registration
Network disconnection	Local communication continues where a MANET path exists; external communication resumes when connectivity returns
Limitation	Complete network isolation can prevent communication with external hospitals or control centres

When is Mobile IP unsuitable?

Mobile IP alone is unsuitable when no reachable Home Agent, Foreign Agent or supporting IP network is available. In such situations, direct MANET communication is more appropriate for local emergency communication.

Conclusion

Before selecting Mobile IP and a routing protocol for a highly mobile wireless environment, network designers should analyze network architecture, node mobility, route availability, communication delay, packet delivery ratio, routing overhead and network failures.

The proposed hybrid architecture combines AODV routing for local communication and Mobile IP for external network connectivity. This provides a flexible approach for emergency communication during natural disasters.